

Project 4 Report

Team name

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(CS 2430, section 502)

Programming Project 4 – Spring 2026

The main question is what the probability of landing on certain squares is in Monopoly | and if the jail strategy will change the probability of landing on certain squares | we used computer simulation because calculating exact probabilities would be too complicated.

Monopoly has many random elements like dice rolls | Chance cards | Community Chest cards | and jail rules.

By running millions of simulated turns, we can see which squares players land on most often.

The board is represented as 40 squares | numbered from 0 to 39 | with GO at 0. The deck is represented as 32 cards with 16 community chests and the rest being chanced.

The player state is for where the player is positioned and if they are in jail

Strategy A is getting out of jail as fast as possible by either using a get out of jail card or paying \$50 to get out, while strategy B is to get doubles to get out or pay \$50 once your three turns were up and you stayed in jail longer.

The deck is randomly sorted, and a card is discarded when used.

When there is no more to use, the deck is shuffled again.

Discussion

As n gets bigger | the closer it gets to the expected average | so if a specific square | like Illinois Avenue | is expected to be landed on frequently due to its position relative to Jail or Chance cards | a higher n ensures that the simulation provides a statistically significant representation of that reality | rather than a result skewed by the variance of a small sample size

For strategy A | for 1000 turns go did have a very from 2.5% to 3.7% but when it went to the one million turns | it was more consistent.

conclusion

Jail was the #1 most landed square for both strategies. Illinois Avenue was always #2.

The top 5 also included GO | New York Avenue | and B&O Railroad. After 100 | 000 turns | the results became very stable.

At 1 | 000 | 000 turns | all 10 runs gave almost identical results.

Since there are only two get-out-of-jail cards, other people will be forced to keep rolling until they get a double or three turns to get out of jail.

Questions

As n got larger | the percentage difference got smaller | so instead of a 1.2% change it went to 0.06%.

After 100 | 000 runs | the variance dropped to 0.1%. At one million n the noticeable difference was that B landed in jail almost twice as much.

There was a 0.2% drop in the number of people staying in jail longer. For squares near the jail.

For multiple people, B would change since there is a limited amount of get out of jail card and more people will stay in jail longer.

RESULTS (top 5, more are in output)

Strategy A

n = 1,000 Turns

Rank	Square Name	Avg %
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1	Jail	6.0000%
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2	Illinois Avenue	3.3500%
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3	Reading Railroad	3.1700%
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4	GO	3.0000%
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5	New York Avenue	2.9900%
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n = 10,000 Turns

Rank	Square Name	Avg %
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1	Jail	6.3450%
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2	Illinois Avenue	3.1820%
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3	New York Avenue	3.1300%
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4	B&O Railroad	3.0310%
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5	Reading Railroad	2.9960%
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n = 100,000 Turns

Rank	Square Name	Avg %
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1	Jail	6.1971%
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2	Illinois Avenue	3.1490%
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3	New York Avenue	3.0838%
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4	GO	3.0807%
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5	B&O Railroad	3.0659%
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n = 1 | 000 | 000 Turns

Rank	Square Name	Avg %
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1	Jail	6.1958%
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2	Illinois Avenue	3.1696%
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3	New York Avenue	3.0798%
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4	GO	3.0773%
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5	B&O Railroad	3.0659%
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Strategy B

n = 1,000 Turns

Rank	Square Name	Avg %
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1	Jail	11.9600%
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2 | New York Avenue | 3.2100%
3 | B&O Railroad | 3.1800%
4 | Pennsylvania Railroad | 3.0300%
5 | Illinois Avenue | 2.9800%

n = 10,000 Turns

Rank | Square Name | Avg %
1 | Jail | 11.1200%
2 | Illinois Avenue | 2.9850%
3 | B&O Railroad | 2.9620%
4 | GO | 2.9190%
5 | New York Avenue | 2.8550%

n = 100,000 Turns

Rank | Square Name | Avg %
1 | Jail | 11.1325%
2 | Illinois Avenue | 2.9760%
3 | GO | 2.9137%
4 | B&O Railroad | 2.8923%
5 | New York Avenue | 2.8422%

n = 1 | 000 | 000 Turns

Rank | Square Name | Avg %
1 | Jail | 11.1040%
2 | Illinois Avenue | 2.9977%
3 | GO | 2.9202%
4 | B&O Railroad | 2.8827%
5 | Free Parking | 2.8258%